

Science Ahead

An International Lower Secondary Science Course

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CHAPTER 2
Food and Digestion



What happens to food that enters the body?

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What Will You Learn in This Chapter?

- Identify the different types of nutrients and their functions.
- Identify what makes a balanced diet and understand the effects of malnutrition.
- Understand how food is digested and the role of enzymes in digestion.
- Recognise the organs of the alimentary canal and their functions.

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2.1 Nutrients in Food

2.2 A Balanced Diet

2.3 Digestion

2.4 The Human Digestive System

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2.1 Nutrients in Food

At the end of this section, you should be able to answer the following questions:

What is the importance of food?

What are the different types of nutrients in food?

What is energy value?

Why is water important?

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Food and its functions

What kinds of food do you eat every day?

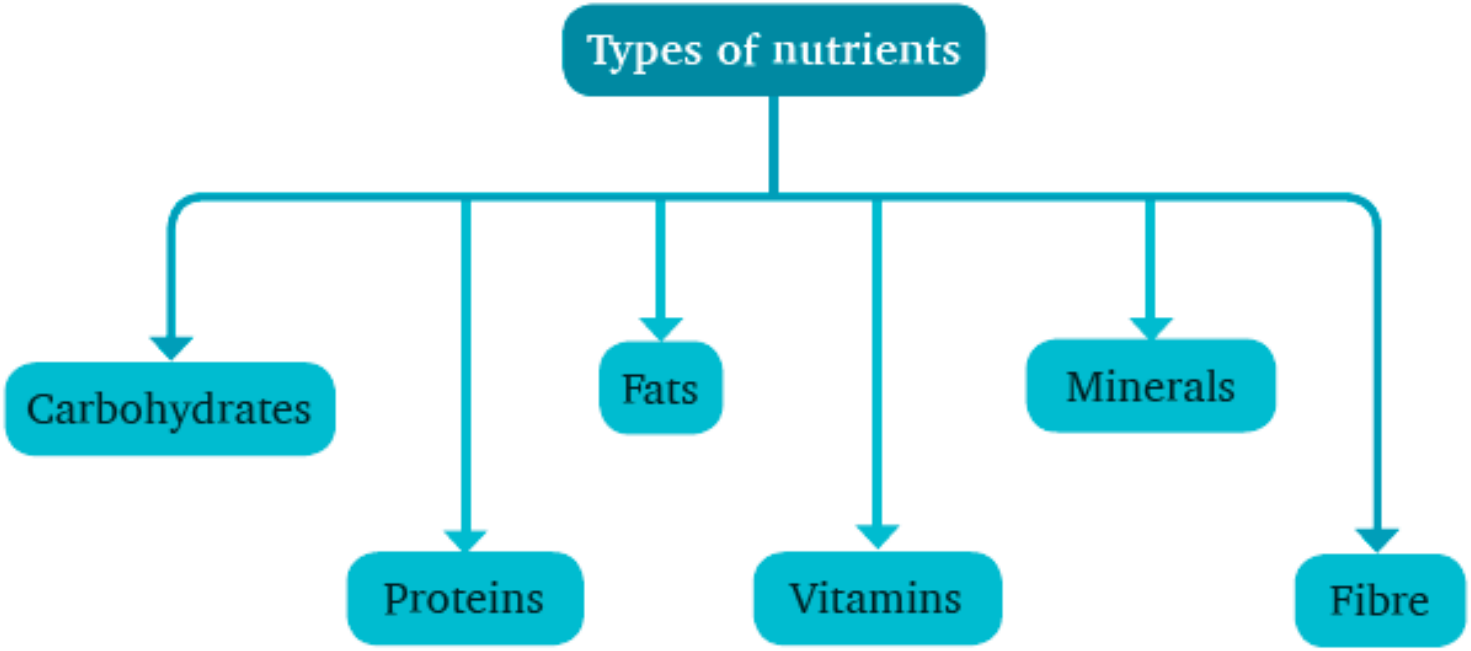
Food provides us with

- **energy** to do work and carry out life processes;
- materials needed for the **growth** of new cells and tissues;
- materials needed for the **repair** of damaged cells and tissues;
- substances that our body needs to **maintain** good health and resist diseases.

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What are the different types of nutrients in food?

Food stores energy and contains essential nutrients that our body needs.



Water is also essential for our body.

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Carbohydrates

- Carbohydrates include **starch** and **sugars**.
- Carbohydrates supply **energy**.
- Examples of carbohydrates are rice, noodles and bread.



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Starch







Which foods in these meals are rich in starch?

- Rice, noodles, bread, potatoes, cereals, yam, cassava and grains are rich in **starch**.

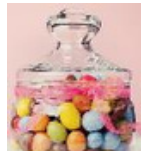
How do we know if the food or drink contains starch?

- **Iodine solution** is used to test for the presence of starch in food.
- The solution is **brown** and turns **blue-black** when starch is present.



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Sugars



- Sugars are found in fruit, candies, cakes and soft drinks.

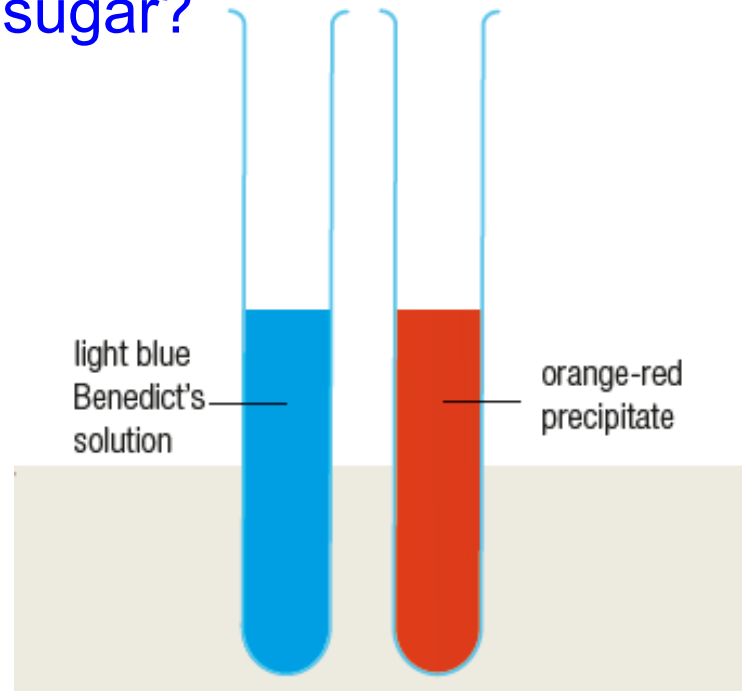


Excess carbohydrates in our body are converted into fats and stored in our body.

How do we know if the food or drink contains sugar?

- **Benedict's solution** is used to test for the presence of **glucose**, a simple sugar.

- The solution is blue. When it is mixed with glucose and heated, an orange-red precipitate is formed.



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Proteins

Name foods that are rich in proteins.

- Foods that are rich in proteins include fish, meat, eggs, cheese, nuts, beans and milk.
- The main function of proteins is to make cells, tissues, enzymes and hormones.
- Proteins also supply us with **energy**.
- Proteins are essential for the **growth** and **repair** of worn-out parts of the body.
- Growing children need plenty of proteins.



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Fats



Name foods rich in fats.

- **Fats from animals** can be found in butter, ghee, cheese, cream, milk, fatty meat and fish.
- **Fats from plants** can be found in nuts, margarine and cooking oils.





- Fats can be **stored** in our body.
- Fats serve as a supply of **energy** when food consumption is low.
- Fats keep us **warm** and absorb nutrients such as **fat-soluble vitamins**.

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Vitamins

Why should we eat vegetables?

- Vegetables contain nutrients called **vitamins**.
- Small amounts of vitamins are needed for good health and growth.
- Some vitamins are soluble in **water** and some are soluble in **fat**.
- **Water-soluble** vitamins such as vitamins B1 and C should be taken regularly because they are not stored in the body.
- **Fat-soluble** vitamins such as vitamins A, D, E and K are stored in fat tissues.
- Excess vitamins in the body can be **toxic**.

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Vitamins (cont'd)

- Different vitamins have different **functions** and come from different **sources**.

- A **lack** of vitamins can lead to health problems and diseases.

Vitamin	Functions	Deficiency problems	Sources
A	• For maintaining healthy skin • For maintaining good eyesight	• Dry and scaly skin • Night blindness	• Carrots, pumpkins, green vegetables, eggs, fish, dairy products

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Vitamins (cont'd)

Vitamin	Functions	Deficiency problems	Sources
B1	• For releasing energy from carbohydrates • For proper functioning of the heart and nervous system	• Beriberi (a disease affecting the nerves and muscles)	• Yeast, wholegrain cereals, peas
C	• For healthy teeth and gums • For absorption of iron (a type of mineral)	• Scurvy (a disease that can lead to swollen and bleeding gums)	• Citrus fruits (e.g. oranges, lemons, limes)

Vitamins (cont'd)

Vitamin	Functions	Deficiency problems	Sources
C	• For healing of wounds and preventing infections • For formation of collagen (a protein in connective tissues) to hold cells together	• Lowers the ability of the body to fight infections • Easy bruising  Mouth of a person suffering from scurvy	• Tomatoes, pawpaw, guavas, green vegetables

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Vitamins (cont'd)

Vitamin	Functions	Deficiency problems	Sources
D	• For growth and development of bones and teeth	• Rickets (a disease that can lead to stunted growth and softening of bones and teeth)  A child suffering from rickets	• Dairy products, eggs, liver • Produced when sunlight falls on our skin.

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Vitamins (cont'd)

Vitamin	Functions	Deficiency problems	Sources
E	• For protecting cells and vital tissues	• Muscle weakness • Difficulty in walking or coordination	• Vegetable oil, wheat germ, nuts, egg yolk
K	• For clotting of blood	• Easy bruising • Prolonged bleeding	• Vegetables, liver • Bacteria in the gut

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Minerals (cont'd)

- We need tiny amounts of minerals for proper **development** and each mineral has a different function.

Mineral	Functions	Deficiency problems	Food sources
Calcium	• For building strong bones and teeth • For muscle growth and contraction • For clotting of blood	• Brittle teeth and tooth decay • Brittle and thin bones that fracture easily	Dairy products, green vegetables, eggs

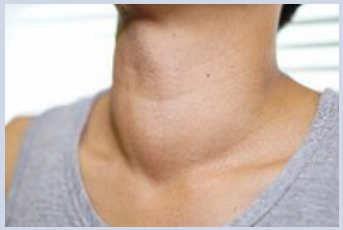
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Minerals (cont'd)

Mineral	Functions	Deficiency problems	Food sources
Iron	• For formation of haemoglobin (the red pigment in red blood cells)	• Anaemia (causes tiredness and lack of energy)	Liver, meat, eggs, green vegetables
Sodium	• For regulating our body fluids	• Fatigue • Muscle cramps • Confusion	Common salt (sodium chloride)

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Minerals (cont'd)

Mineral	Functions	Deficiency problems	Food sources
Iodine	• For formation of hormones in the thyroid gland (an organ in the neck)	• Goitre (the swelling of the thyroid gland)  Person suffering from goitre	Seafood, seaweed, green vegetables, table salt (which often has iodine added to it)

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Fibre

- Fibre or roughage is made up of **cellulose** that cannot be broken down by our body.
- It passes through and leaves our body in the form of **faeces**.
- The right amount of fibre in our meals keeps the intestines healthy and working properly.

Can you name foods that are rich in fibre?

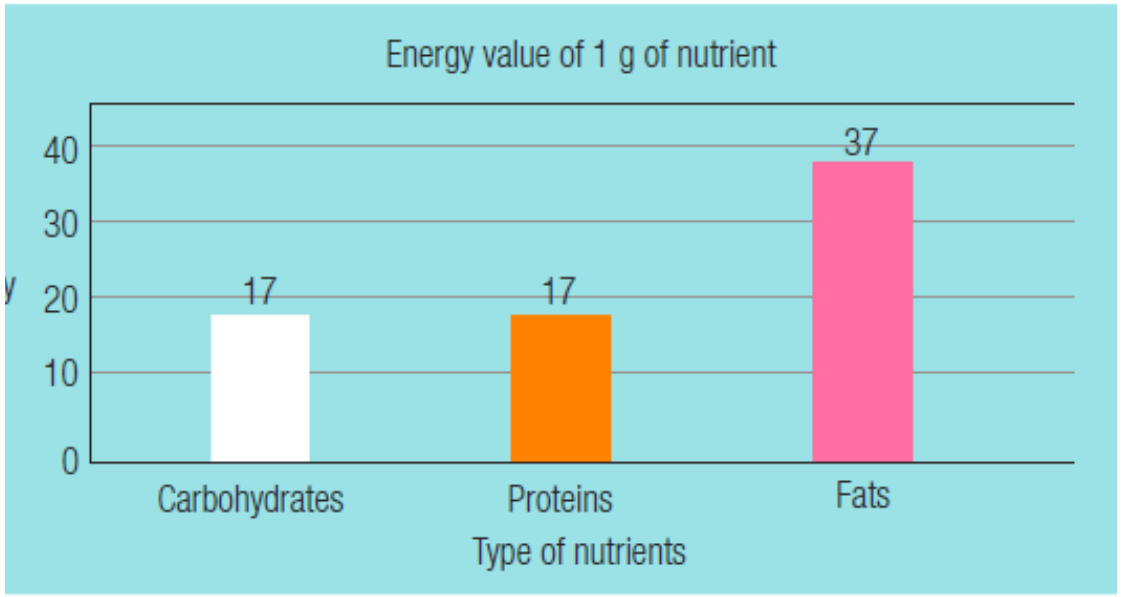
Fruits, vegetables, brown rice and wholemeal bread are examples of foods that are rich in fibre.



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Energy values of different nutrients

Energy value is the amount of energy stored in food.



Energy values of carbohydrates, proteins and fats

- **Carbohydrates** do not have the highest energy value, but they are the first source of energy the body uses.
- **Fats** are used when the carbohydrates in our body are used up.
- **Proteins** are used to supply the body with energy when carbohydrates and fats are not available.
- **Muscles** will break down if proteins are used as energy source.

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Why is water important?

- Water makes up about 60% to 70% of our **body mass**.
- Water is needed for **breaking down** food, **dissolving** nutrients and **transporting** them around the body.
- Water is the main component of **cells** and **blood plasma**.
- **Chemical reactions** can only take place in the **cytoplasm** of cells that contains enough water.
- The body loses water through **sweat**, **urine** and the **air** we breathe out. The water lost needs to be **replaced.**
- An average person needs **six to eight glasses** of water each day. Eating food that is high in **water content**, such as juicy fruits also helps to obtain water.

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Key Points

- Food provides us with
  - energy to do work and carry out life processes;
  - materials needed for the growth of new cells and tissues;
  - materials needed for the repair of damaged cells and tissues;
  - substances that the body needs to maintain good health and resist diseases.
- Most food contains a mix of different types of nutrients such as carbohydrates, proteins, fats, vitamins, minerals and fibre.
- Carbohydrates include starch and sugars. They supply most of the energy needed by the body.

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Key Points (cont'd)

- Proteins are needed to make cells, tissues, enzymes and hormones.
- Fats serve as a reserve supply of energy and keep us warm.
- Vitamins are needed in small amounts by the body for good health and growth.
- Minerals are needed by the body in tiny amounts for growth and development.
- Fibre is made up of cellulose that cannot be broken down by the body.
- The amount of energy stored in food is known as the energy value.
- Water is needed for various life processes and chemical reactions in the body, such as digestion and transportation of nutrients.

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2.1 Nutrients in Food

2.2 A Balanced Diet

2.3 Digestion

2.4 The Human Digestive System

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What is a varied diet?

Do you eat a variety of food every day?

How much do you eat at each meal?

- **Diet** refers to the food and drink a person usually consumes.
- A varied diet contains different types of food that supplies us with **energy** and **nutrients** needed by our body.

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2.2 A Balanced Diet

At the end of this section, you should be able to answer the following questions:

What is a balanced diet?

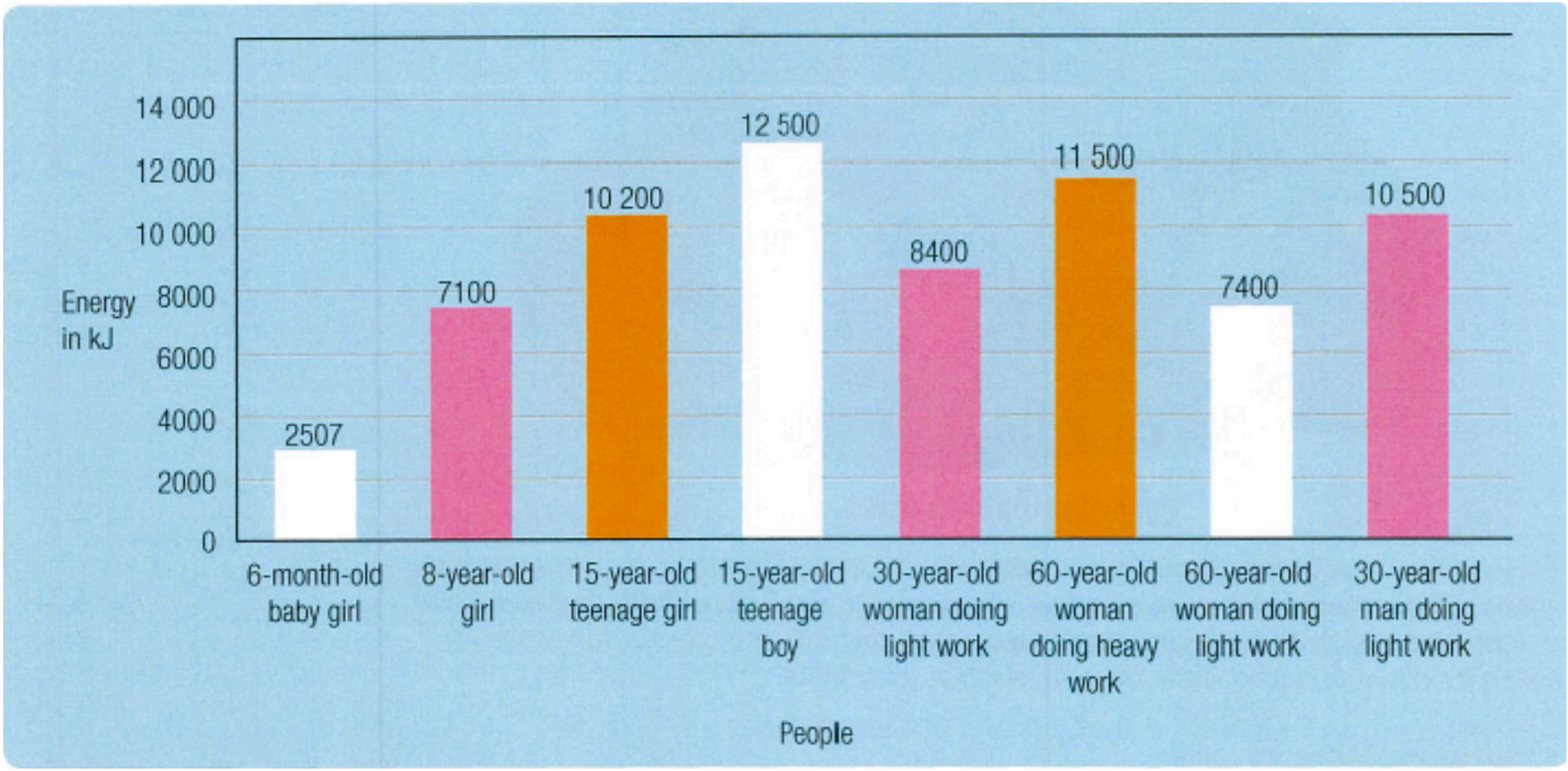
What is malnutrition?

What happens if our body takes in more energy than what is used up?

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Different people have different energy needs

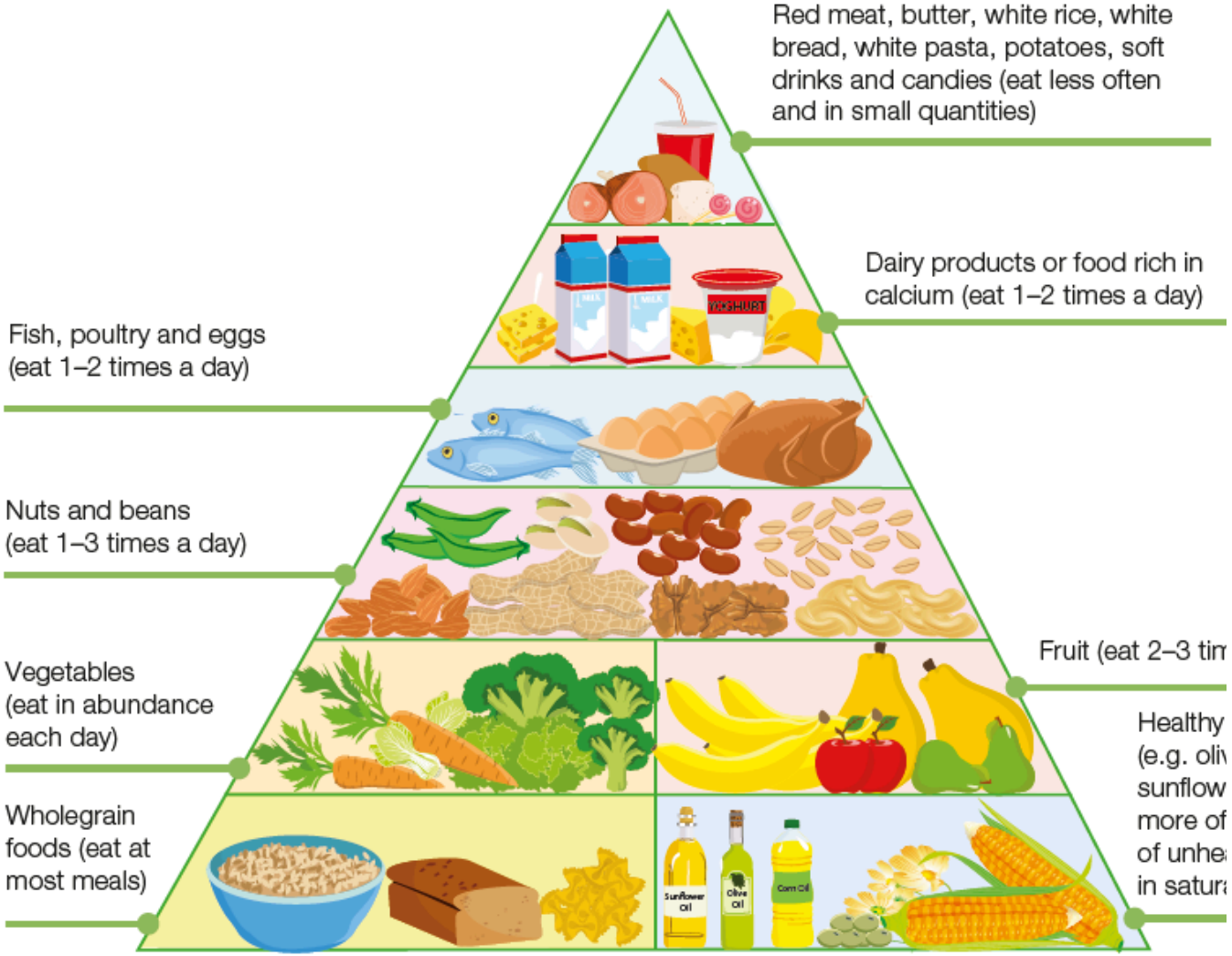
How much energy do you need daily?



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What is a balanced diet?

A **balanced diet** contains the right amounts of carbohydrates, proteins, fats, vitamins, minerals, fibre and water to meet the nutritional and energy needs of our body.



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Malnutrition

- **Malnutrition** refers to too much or too little intake of energy or nutrients.
- Carbohydrates that are not used up are stored in the body as **fats**, which can lead to **obesity**.
- Obesity is linked to **heart disease**, **diabetes** and **high blood pressure**.
- High blood pressure can lead to diseases such as **stroke** and **kidney failure**.

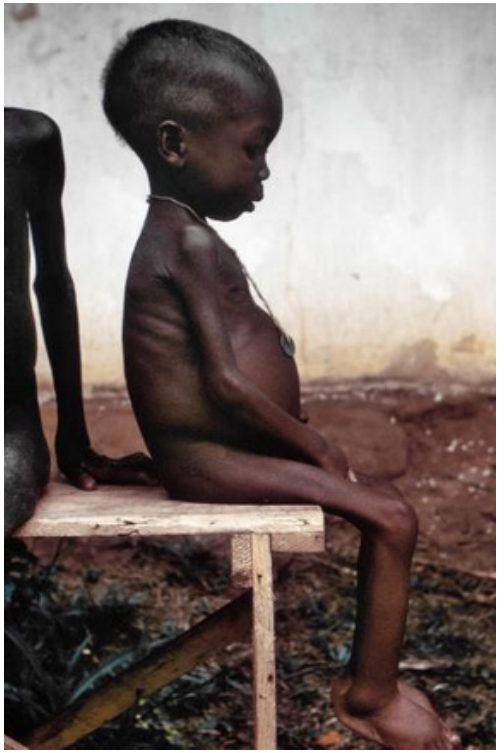


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Malnutrition (cont'd)



- Not eating enough can lead to **deficiency** problems.
- **Anorexia** is a condition in which a person refuses to eat or eats very little for fear of gaining weight.
- People suffering from anorexia **starve** themselves on purpose and **exercise excessively**. They suffer from weakness, fatigue and deficiency problems.
- A severe lack of proteins in the diet can result in a disease called **kwashiorkor** in children. This is common in areas with famine.



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Key Points

- The energy needs of people are different.
- A balanced diet contains the right amount of carbohydrates, proteins, fats, vitamins, minerals, fibre and water to meet the needs of our body.
- Malnutrition, which refers to too much or too little intake of energy and nutrients, can arise if our diet is not balanced.
- If our body takes in more energy than is used up, it stores the extra energy as fat. Too much fat can lead to obesity and related problems.
- Not eating enough can lead to deficiency problems.

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2.1 Nutrients in Food

2.2 A Balanced Diet

2.3 Digestion

2.4 The Human Digestive System

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What is digestion?

Imagine you have just eaten a slice of cake. Does the cake still look the same in your mouth and stomach?

- The food that we eat is broken down into smaller pieces so that it can be absorbed by the body.
- Nutrients in food need to reach the the cells in our body before they can be used.



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At the end of this section, you should be able to answer the following questions:

What is digestion?

What is the difference between physical digestion and chemical digestion?

Why must food be digested?

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What is digestion? (cont'd)

- Large and **insoluble** molecules, such as starch, proteins and fats cannot pass through the **cell membranes** into our cells.
- The **molecule** is the smallest particle in a substance that has the chemical properties of that substance.
- **Digestion** is the process of breaking down food into smaller, soluble molecules. Digestion can be **physical** or **chemical**.

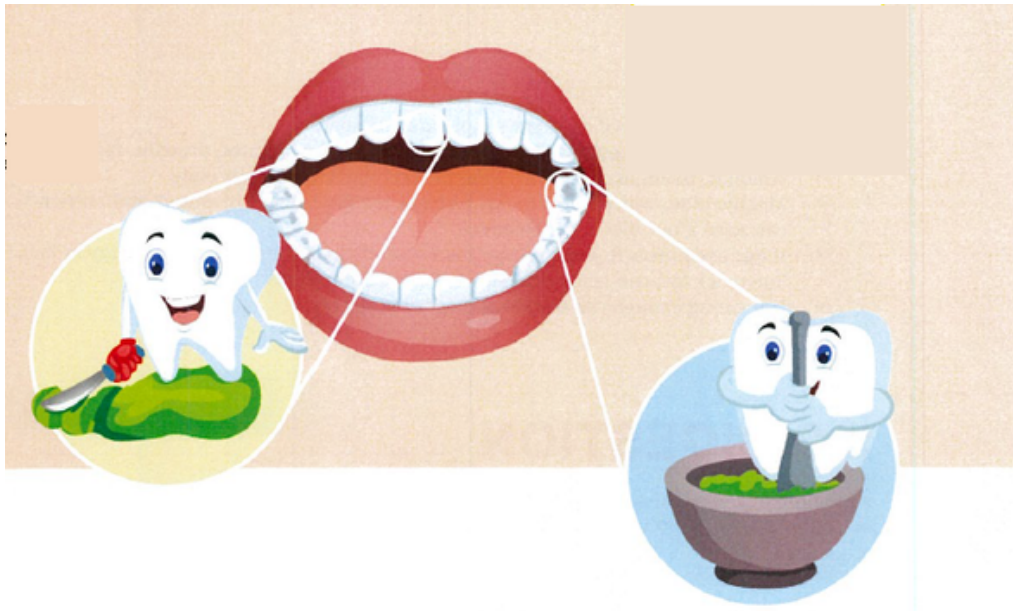
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Physical digestion

What happens to food in physical digestion?



- In physical digestion, big pieces of food are **broken down** into smaller pieces.
- There is **no change** in the molecules, so no chemical reaction takes place.
- **Incisors** cut and slice food into small pieces.
- **Molar teeth** crush small pieces of food.

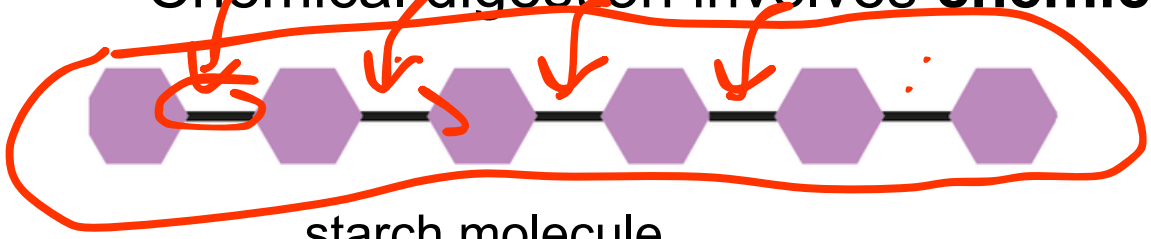


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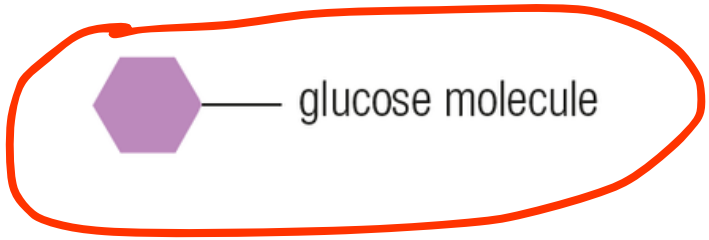
Chemical digestion

What happens in chemical digestion?

- Large food molecules are **broken down** into smaller molecules, forming a **new molecule**.
- Chemical digestion involves **chemical reactions**.



starch molecule

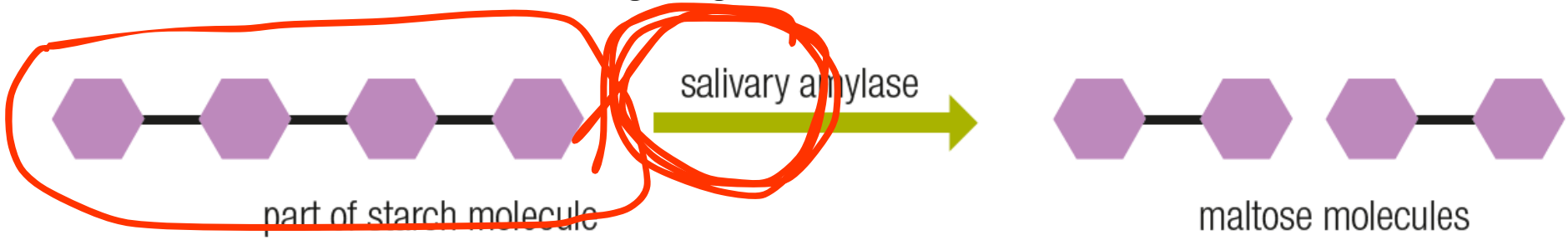


- **Glucose molecules** are small enough to pass through the wall of the small intestine into the bloodstream.
- The **blood** carries the glucose molecules to different parts of the body.

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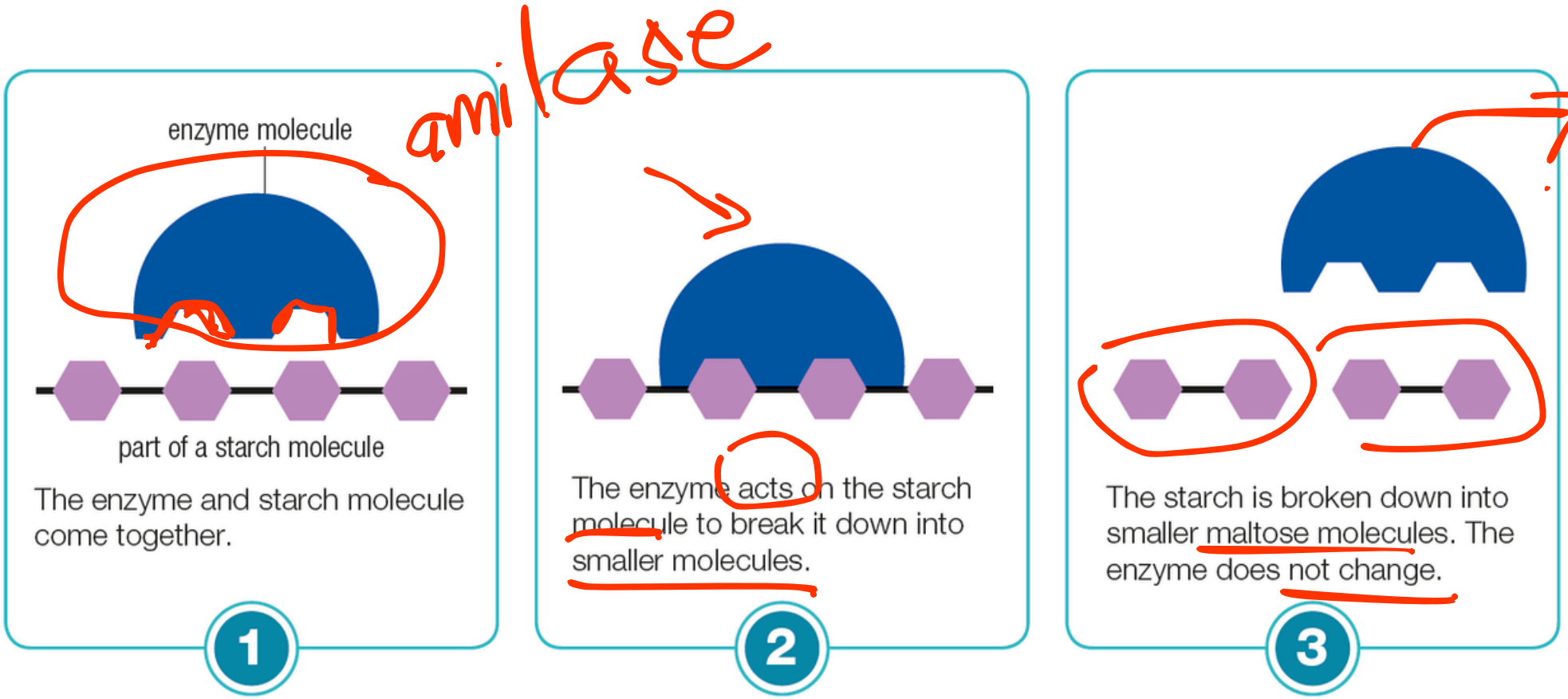
Enzymes

- Enzymes are special proteins produced by the body that **speed up** chemical digestion.
- Substances that speed up specific chemical reactions are called **catalysts.**
- Enzymes are **biological catalysts**.
- Enzymes that speed up digestion are **called digestive enzymes**.
- Different digestive enzymes break down nutrients into **smaller** and **soluble** molecules during digestion.



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Action of enzyme on a starch molecule



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Key Points

- Digestion is the breakdown of large insoluble molecules into smaller soluble ones.
- Food must be digested because most food molecules are too large or insoluble to pass through the cell membranes into cells.
- Physical digestion is the breakdown of big pieces of food into smaller pieces. It does not involve chemical reactions.
- Chemical digestion is the breakdown of large food molecules into smaller ones. It involves chemical reactions.
- Enzymes are biological catalysts produced by the body that speed up certain chemical reactions in the body.
- Digestive enzymes help in digestion. For example, the digestive enzyme salivary amylase breaks down starch into maltose.

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2.1 Nutrients in Food

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2.3 Digestion

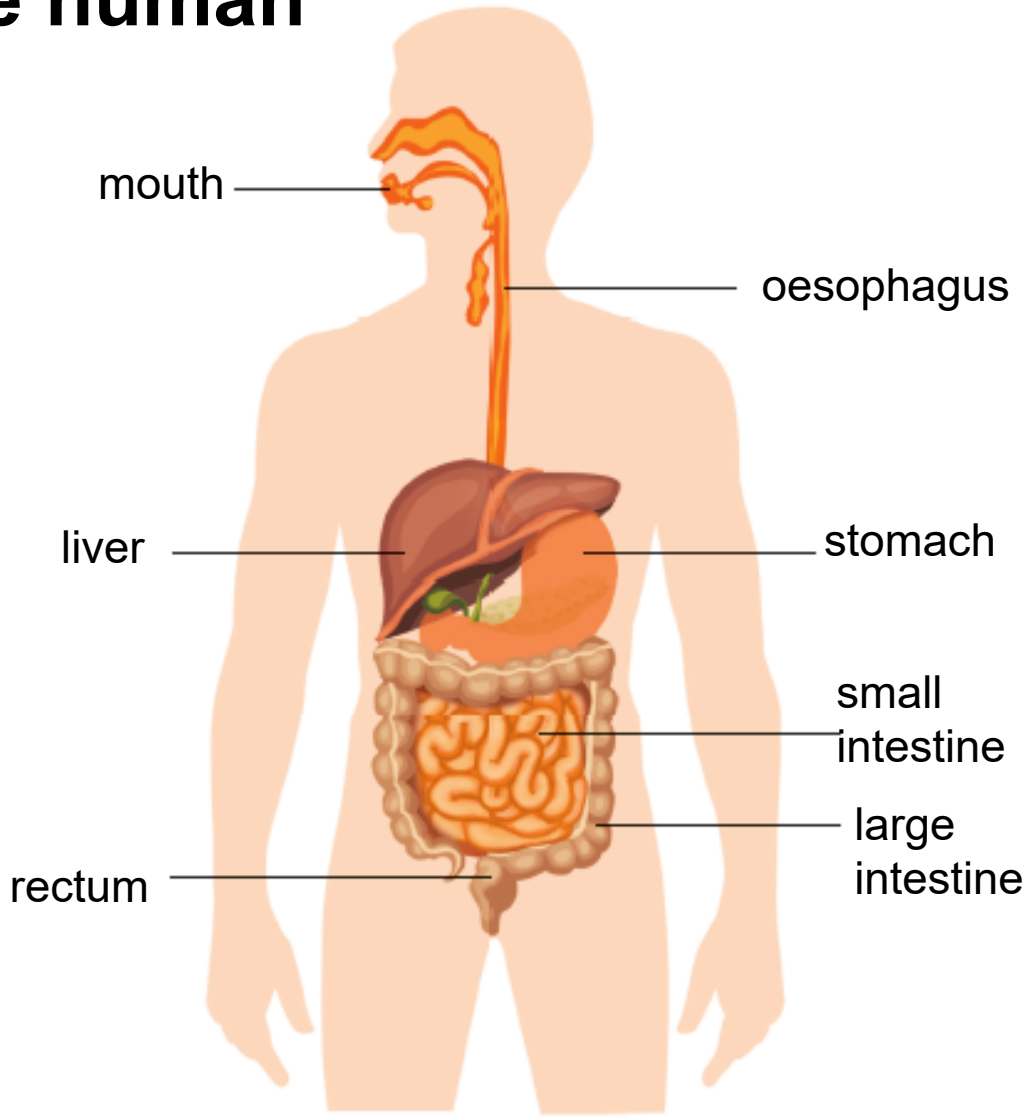
2.4 The Human Digestive System

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What are the parts of the human digestive system?

- The human digestive system is made up of the **alimentary canal** or **gut**, the **liver**, the **pancreas**, and the **gall bladder**.

- The alimentary canal is a continuous tube that begins in the **mouth** and ends at the **anus**.



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2.4 The Human Digestive System

At the end of this section, you should be able to answer the following questions:

What are the organs in the human digestive system?

What happens to food in each part of the alimentary canal?

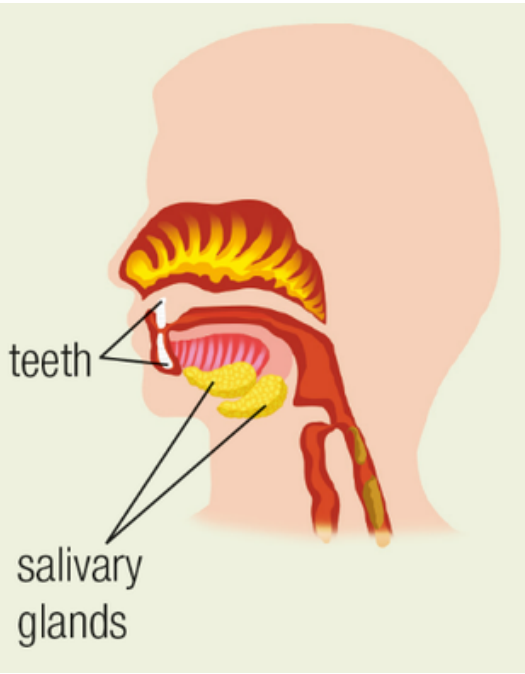
Why must food be digested?

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Mouth

Digestion of food starts in the mouth.

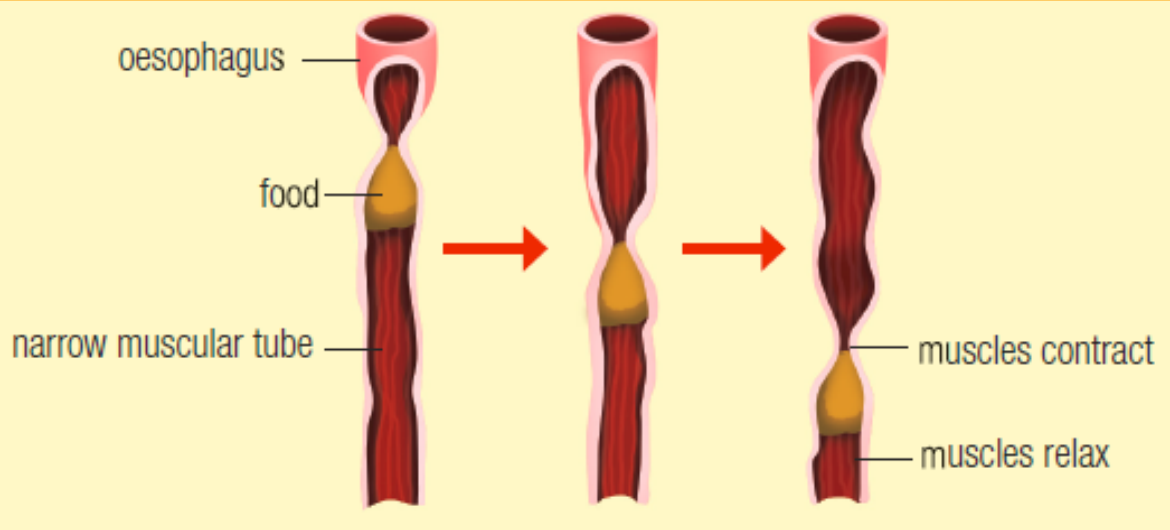
Parts	Function
Teeth	The teeth break up food into smaller pieces. This increases the **surface area** of the food for the **digestive enzymes** to act on.
Salivary glands	The salivary glands produce saliva in or around the mouth. The saliva contains the digestive enzyme **salivary amylase**. Salivary amylase breaks down large **starch** molecules into smaller **maltose** molecules.
Tongue	The tongue rolls the partially digested food into small balls. These are then swallowed and move into the **oesophagus.**



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Oesophagus

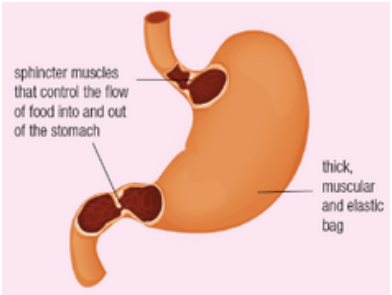
- The oesophagus is a **narrow tube** with **strong muscles** in its wall.
- The muscles contract and relax, producing wave-like movements called **peristalsis**, which pushes the food along the oesophagus to the stomach.
- No digestive enzymes are produced in the oesophagus.



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Stomach

- The stomach is a **muscular** organ with **glands** in the wall that secrete **gastric juice**.
- Gastric juice contains the following:
  - Digestive enzymes called **proteases**, which digest long **protein chains** into shorter protein chains
  - **Hydrochloric acid**, which helps proteases work and kills **unwanted bacteria**
- **Peristalsis** in the stomach **churns** the food and mixes it with **gastric juice**.
- Food turns into a semi-liquid mass called **chyme** and is released in the small intestine in small amounts at different time intervals.



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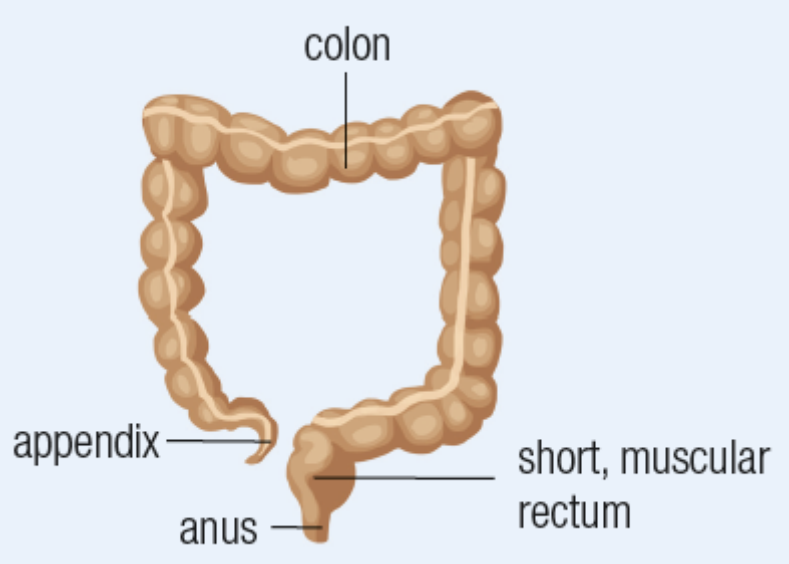
Small intestine

- The small intestine is a **long muscular tube**.
- These are the functions of the small intestine:
  - It **contracts** and **releases** to mix the food with more digestive juices to help digestion.
  - It **absorbs** digested food, and some **vitamins** and **minerals** into the **bloodstream**.
- Digestion of food ends in the small intestine.
- Excess water, remaining minerals and undigested food such as fibre, are passed to the **large intestine** by peristalsis.

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Large intestine

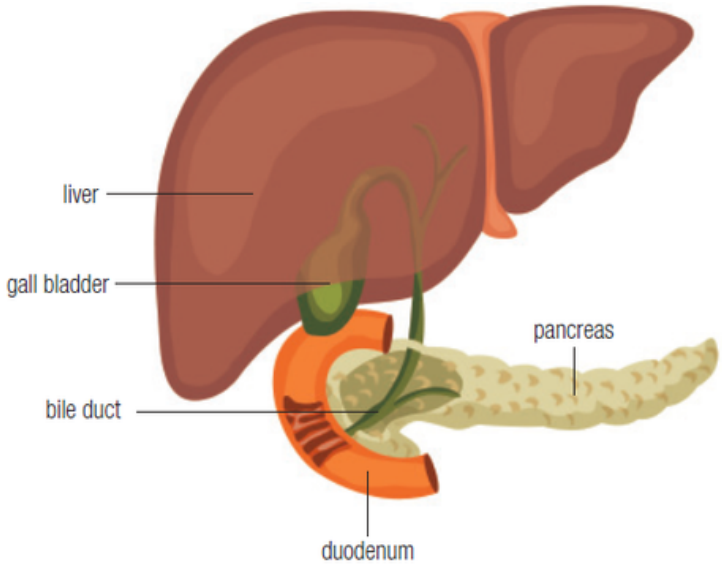
Parts	Functions
Colon	Absorbs **excess water**, and remaining **minerals** and **vitamins** produced by bacteria in the large intestine, such as **vitamin K**
Rectum	Stores undigested contents as **faeces**
Anus	**Expels** faeces when the ring-like muscles between the rectum and anus relax



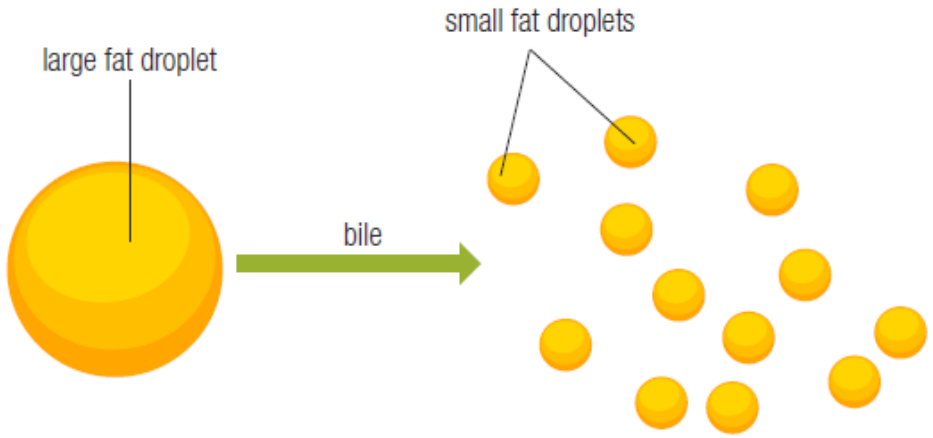
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What happens in the small intestine?

Food from the stomach enters the **duodenum** a little at a time.



Bile produced by the **liver** is squirted through the **bile duct** onto the food. It breaks up large **fat droplets** into smaller fat droplets. Bile is stored in the **gall bladder**.



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What happens in the small intestine? (cont'd)

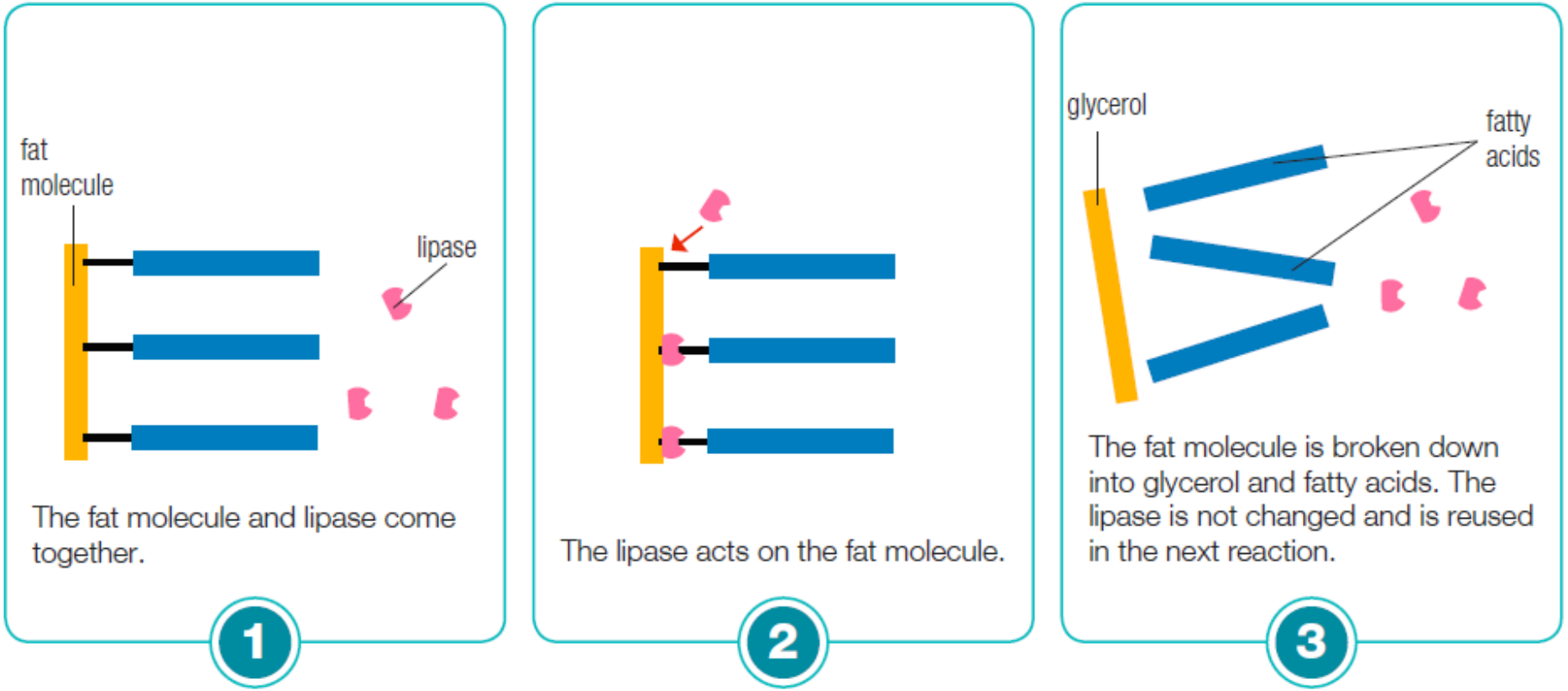
Pancreatic juice from the pancreas and **intestinal juice** from the small intestine contain the following enzymes to digest the different nutrients in the small intestine:

- **Lipase**, which breaks down fats into simple units of **fatty acids** and **glycerol**
- **Proteases,** which break down shortened **protein chains** into small units of **amino acids**
- **Carbohydrases,** including **amylase** from the pancreas, which break down partially digested **carbohydrates** into **simple sugars**

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What happens in the small intestine? (cont'd)

Lipase breaks down fat molecules.



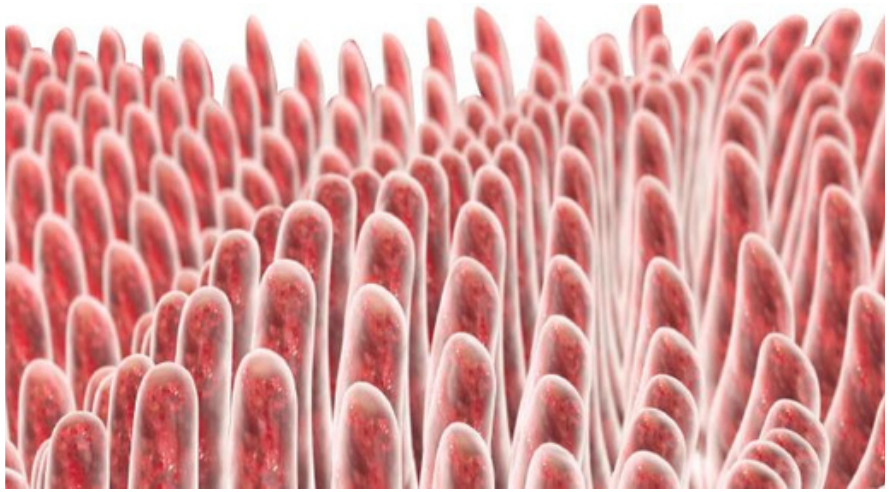
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What happens in the small intestine? (cont'd)

- Digested food molecules are absorbed into the bloodstream through the **villi**.
- Villi are tiny **finger-like projections** on the inner wall of the small intestine.
- They increase the **surface area** for the absorption of digested food.

Some vitamins and minerals that are small and do not need to be digested are absorbed by the small intestine.

These are **transported** to the various parts of the body by the **blood**.



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Key Points

- The human digestive system consists of the alimentary canal, the liver, the pancreas and the gall bladder. Digestion of food starts in the mouth and ends in the small intestine.

- Salivary amylase in the mouth digests starch into maltose.

- The oesophagus pushes food down from the mouth into the stomach by contracting and relaxing its muscular walls. This movement is called peristalsis.

- Proteases in the gastric juice produced by the stomach digest long protein chains into shorter protein chains. Hydrochloric acid in the gastric juice activates the proteases and kills bacteria.

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Key Points (cont'd)

- Bile produced by the liver breaks up large fat droplets into smaller fat droplets for the digestive enzymes in the small intestine to act on.

- Pancreatic and intestinal juices in the small intestine contain digestive enzymes which digest

  - fats into fatty acids and glycerol;
  - proteins into amino acids;
  - starch into maltose;
  - maltose into simple sugars.

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Key Points (cont'd)

- Digested food, and some vitamins and minerals are absorbed into the bloodstream through the wall of the small intestine.
- Excess water, the remaining minerals and undigested food such as fibre are passed into large intestine.
- Excess water, the remaining minerals and vitamins produced by bacteria in the large intestine are absorbed into the bloodstream. The undigested food is stored in the rectum and passed out as faeces through the anus.

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Chapter review and extension

- **Reflection** (Workbook page 13)
- **Review Questions** (Student Book pages 34–35)
- **Revision Worksheet 2** (Workbook pages 14–17)
- **Think-tank** (Student Book pages 36–37)

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